INCH-POUND

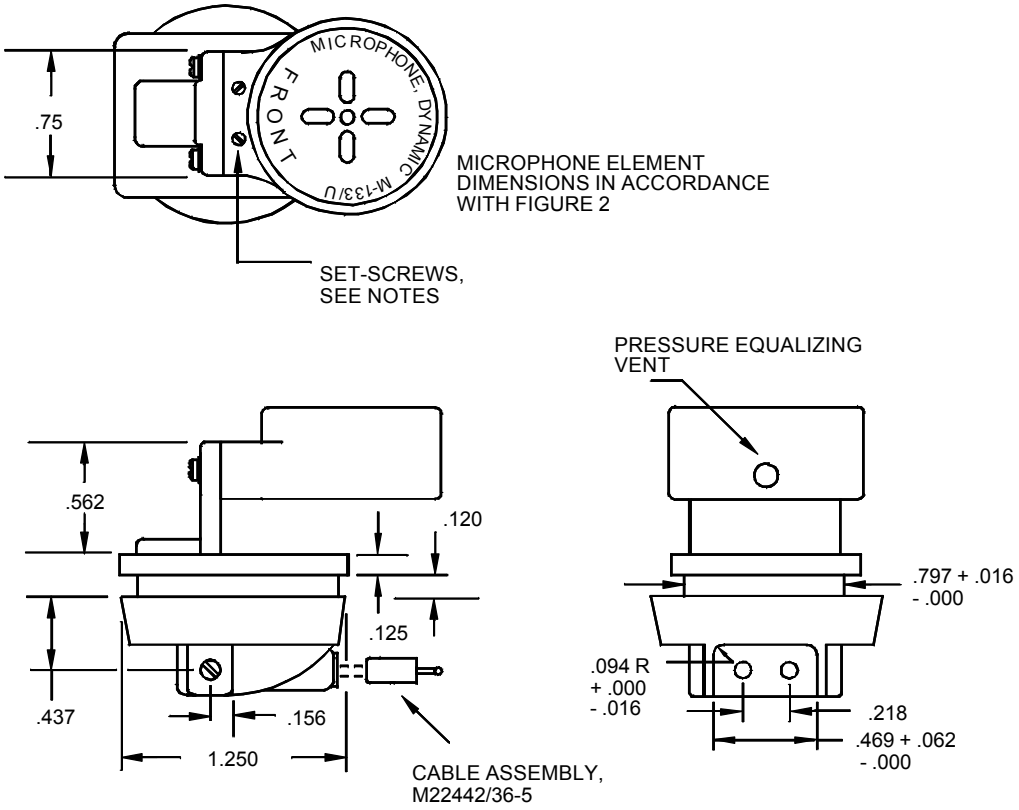
MIL-PRF-26542/9B
30 May 1997
SUPERSEDING
MIL-M-26542/9A
29 March 1985

PERFORMANCE SPECIFICATION SHEET

MICROPHONE ASSEMBLY AND MICROPHONE, DYNAMIC,
M-133/U AND M26542/9-01

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.

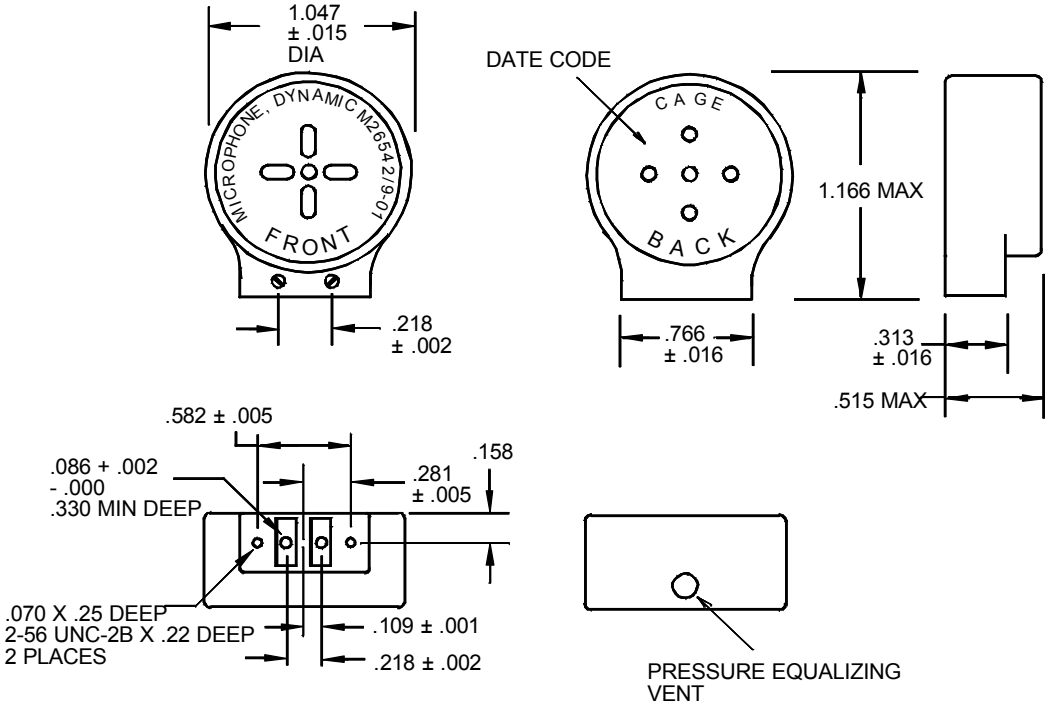


NOTES:
1. Dimensions are in inches. Tolerance is ±.015 inches, unless otherwise specified.
2. Quantity and configuration of sound apertures is optional.
3. Marking location on surface shown optional.
4. Set screws shall secure the element to the bracket securely, of type slotted for consistency with established US Army field-repair procedures and tooling, and shall be recessed.

FIGURE 1. Microphone assembly, M-133/U .

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NOTES:

1. Dimensions are in inches. Tolerance is ±.015, unless otherwise specified.
2. Configuration and number of sound apertures is optional.
3. Location of marking on surface shown is optional.
4. Set-screws shall enable the unit to mount securely to the associated bracket, and shall be of type slotted, for consistency with US Army field-repair procedures and tooling, and shall be recessed.

FIGURE 2. Microphone, M26542/9-01 .

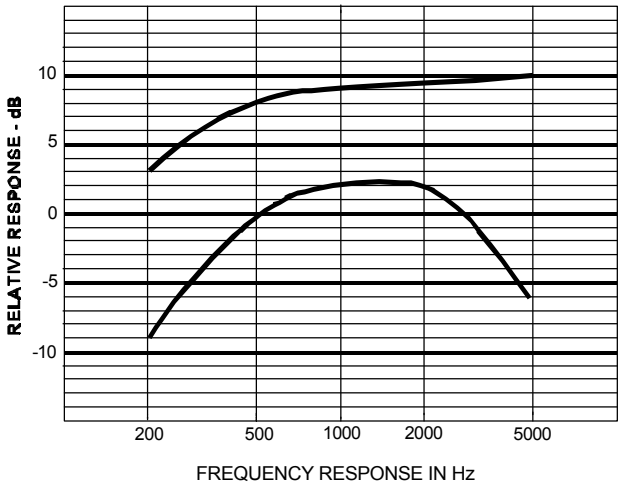


FIGURE 3. Frequency response at ground level .

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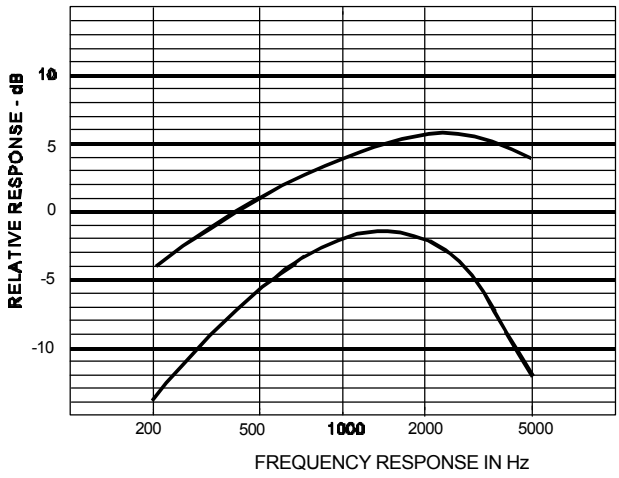


FIGURE 4. Frequency response at 25,000 feet .

REQUIREMENTS:

Component parts:

Microphone element: Shall be in accordance with figure 1 or figure 2, as specified by the part number (see table I).

Bracket: Shall be in accordance with figure 1, for interface to the dimensions of the gas mask assembly. Two (2) screws securing the bracket to the microphone element, type AN500D-6 (3/8 in) 2-56 thread, shall be supplied for interchangeability. The bracket shall also be supplied with suitable hardware to ensure that the microphone element does not become disengaged during normal use, such as lockwashers.

Cable assembly: Shall be Part or Identifying Number (PIN) M22442/36-5, for interchangeability, and to meet the environmental requirements of this specification and the end-item-assembly.

Plug assembly: U-173/U, in accordance with USAF Drawing 57B12662, or electrically and mechanically compatible part meeting the environmental requirements of MIL-PRF-26542.

Performance:

Sensitivity at ground level: 36.90 dB - 42.92 dB (re 1 µV) or 69.98 µV - 139.63 µV with an SPL input of 28 dynes/cm^2 at 1 kHz.

Sensitivity at altitude: To be within 8 dB of measured value at ground level.

Frequency response at ground level: Shall be in accordance with figure 3.

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Frequency response at altitude: Shall be in accordance with figure 4.

Impedance: 4.0 to 6.0 ohms.

Resistive load: 5.0 ohms.

The microphone assembly shall be tested in accordance with the tests listed in table II.

Intended use: This microphone is used primarily in US Army applications requiring the M45 or the M40A1series of gas masks.

TABLE I. Part number designations.

Part or Identifying Number (PIN)	Characteristics
M-133/U	Supplied as shown in figure 1, with the bracket and cable assembly specified.
M26542/9-01	Supplied with the microphone element only.

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TABLE II. Parameter applicability.

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1, 3.4, 3.5	4.5.1	X	X		
Sensitivity at ground level	3.5.1	4.5.2.2	X	X		
Sensitivity at altitude	3.5.2	4.5.2.3	X			
Frequency response at ground level	3.5.3	4.5.2.4	X	X		
Frequency response at altitude	3.5.4	4.5.2.5	X			
Impedance	3.5.5	4.5.3	X	X		
Noise cancellation characteristics	3.5.7.1	4.5.5.1				
Effect of external magnetic field	3.5.8	4.5.6	X			
Stray magnetic field	3.5.9	4.5.7	X			
Linearity	3.5.10	4.5.8	X			
Talk-out	3.5.11	4.5.9				
Dielectric withstanding voltage	3.5.12	4.5.10	X			
Signal-to-noise	3.5.7	4.5.5	X		X	
Distortion	3.5.6	4.5.4	X		X	
Interchangeability	3.5.13	4.5.11				
Group II						
Thermal shock	3.5.14	4.5.12	X			X
Humidity	3.5.15	4.5.13	X			X
Drop	3.5.16	4.5.14	X			X
Pressure equalization	3.5.17	4.5.15	X			X
Explosive decompression	3.5.18	4.5.16				
Salt fog	3.5.19	4.5.17				
Group III						
Vibration	3.5.20	4.5.18	X			X
Bounce	3.5.21	4.5.19	X			X
Altitude	3.5.22	4.5.20	X			X
Moisture barrier seal	3.5.23	4.5.21				
Immersion	3.5.24	4.5.22	X			X
Group IV						
Fungus	3.5.25	4.5.23	X			
Group V						
Gun blast	3.5.26	4.5.24	X			

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, EA, ME
Navy - AS, OS
Air Force - 99
DLA - CC

Preparing activity:
Air Force - 85

(Project 5965-0238-09)

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